



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	123

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	6
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	5

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge Legibility	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mr. Ramesh Joshi

Signature :

Date :

2. i) Case study - An e-commerce platform wants to improve its recommendation system to boost user engagement and sales. The current system suggests products based on popularity, but many users find the recommendations irrelevant. The company plans to implement a personalized system using user behavior, purchase history and browsing patterns while ensuring scalability.

→ (i) Recommending products to new users -

- Immediate strategies -
- onboarding questions - Ask preferences.
- context-based recommendation - location, device, time
- session-based behavior - first few clicks.
- Trending + semi-personalized - popular items within selected category.
- content-based approach -
- Use - product metadata
- match with - user's initial interactions.

(ii) which approach? (CF vs content vs hybrid)

- best choice - hybrid recommendation system.
- why? - collaborative filtering - captures "users like you!" patterns
- works well for mature users.
- content based filtering - use item features.
- Hybrid - best approach.
- Use - CF for personalization, content-based for cold start.
- Ex. new user, active user, old user.

(iii) Preventing bias & ensuring diversity.

- Problem - popularity bias - some products shown repeatedly.
- filter bubble - user sees only narrow category.
- Solutions -
- diversity techniques -

- Re-ranking algorithms - mix-similar + diverse items.
- maximal marginal relevance - balance relevance vs diversity.
- category spread - ensure multiple categories in recommendations.
- bias reduction - Personalize over popular items.
- boost - new items, long-tail products.
- exploration vs exploitation.
- Use - multi-armed bandits.

(iv) Real-time recommendations updates.

- Techniques:
- streaming + event processing - tools kafka / real time pipelines.
- capture clicks views, acts actions instantly.
- session based models - Use RNNs, transformers & sequential models.
- Incremental model updates - update embeddings or scores in near real-time.
- Caching + fast retrieval - Use Redis / vector database.
- Two-tower / embedding models - capture user embedding

(v) Balancing personalization with privacy.

- challenges - using user data vs protecting user trust.
- solutions - data minimization - collect only necessary data
 - avoid sensitive attributes.
- Anonymization - use ids instead of personal info
- federated learning - Train models on-device
 - only share model updates.
- differential privacy - add noise to data.
- Transparency & control - view/edit preferences
 - opt out of tracking.
- compliance - GDPR-like principles
 - even outside EU